

Jan Mazal (Ed.)

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Modelling and Simulation for Autonomous Systems

4th International Conference, MESAS 2017
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Revised Selected Papers



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Editor
Jan Mazal
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Preface

This volume contains the full papers presented at the MESAS 2017 Conference: Modelling and Simulation for Autonomous Systems, held during October 24–26, 2017, in Rome. The initial idea to launch the MESAS project was introduced by the Concept Development and Experimentation Branch at the NATO Modelling and Simulation Centre of Excellence in 2013, namely, by LTCDR Alessandro Cignoni, who wanted to bring together the modelling and simulation and the autonomous systems communities and to collect new ideas for concept development and experimentation in this domain. From that time, the event gathers together — in keynote, regular, poster, and way-ahead sessions — fully recognized experts from different technical communities in the military, in academia, and in industry. The main topical parts of the 2017 edition of MESAS were “M&S of Intelligent Systems – AI, R&D, and Applications,” “Autonomous Systems in Context of Future Warfare and Security – Concepts, Applications, Standards, and Legislation,” and “Future Challenges and Opportunities of Advanced M&S Technology.” The community of interest submitted 38 papers for consideration. Each submission was reviewed by two Program Committee members or selected independent reviewers. The committee decided to accept 33 papers to be presented (in seven sessions) and these papers were also accepted to be included in the conference proceedings. The plenary session and way-ahead included an extra three invited presentations from distinguished experts.

October 2017

Jan Mazal

MESAS 2017 Logo



MESAS 2017 Organizer

NATO Modelling and Simulation Centre of Excellence (NATO M&S COE)

The NATO M&S COE is a recognized international military organization activated by the North Atlantic Council in 2012, and does not fall under the NATO Command Structure. Partnering nations provide funding and personnel for the centre through a memorandum of understanding. The Czech Republic, Italy, and the United States are the contributing nations, as of this publication. The NATO M&S COE supports NATO transformation by improving the networking of NATO and nationally owned M&S systems, promoting cooperation between nations and organizations through the sharing of M&S information, and serving as an international source of expertise.

The NATO M&S COE seeks to be a leading world-class organization, providing the best military expertise in modelling and simulation technology, methodologies, and the development of M&S professionals. Its state-of-the-art facilities can support a wide range of M&S activities including but not limited to: education and training of NATO M&S professionals on M&S concepts and technology with hands-on courses that expose students to the latest simulation software currently used across the alliance; concept development and experimentation using a wide array of software capability and network connections to test and evaluate military doctrinal concepts as well as new simulation interoperability verification; and the same network connectivity that enables the COE to become the focal point for NATO's future distributed simulation environment and services.



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M&S of Intelligent Systems - AI, R&D and Applications

Autonomous Systems in Context of Future Warfare and Security - Concepts, Applications, Standards and Legislation



Operational Scenario Modelling Supporting Unmanned Autonomous Systems Concept Development

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Abstract. There exists a connection between a growing number of Unmanned Autonomous Systems (UAXS) and a number of possible security challenges. The likelihood that swarms of UAXS will be employed in high populated mega-city environments in the long and mid-term exacerbates these future challenges. The problem is two-fold: UAXS are likely to be used as additional capability for friendly forces, but they are also to be considered as possible threat. The challenges that an autonomous system poses depend not only on their physical vulnerability or the security of the networks and data, but also on the interaction with traditional troops and operating procedures to be followed, their levels of autonomy and Command and Control (C2) chain behind their employment. An extensive experimentation is necessary to address some of these challenges in a Concept Development and Experimentation (CD&E) process aimed to envision solutions. Modelling and Simulation (M&S) tools and architecture are essential to speed up this process in a cost effective way. This paper presents how the development of a particular air operational scenario in an urban environment, according to the Simulation Interoperability Standard Organization (SISO) guide using M&S tools, can support the capability development process for future UAXS as well as their countermeasures.

In particular, the M&S architecture was made of a federation of simulators interacting with real C2 systems. Several technologies were exploited, such as High Level Architecture (HLA), for simulation Run-Time Infrastructure; Artificial Intelligence (AI) ad-hoc developed modules, for robotic behavior; C2 – Simulation Interoperability (C2SIMI) language, for interoperation between simulators and C2 systems. The scenario was designed for addressing the challenges and countermeasures in the air domain operations concerning specifically facing an hostile swarm of Unmanned Autonomous Aerial Vehicles (UAaVs) attacking a critical infrastructure in a highly populated city.

Keywords: Unmanned Autonomous Systems · Operational scenario
Modelling and Simulation

1 Introduction

There exists a connection between a growing number of Unmanned Autonomous Systems (UAXS) and a number of possible security challenges [1]. The likelihood that swarms of UAXS will be employed in high populated mega-city environments in the long and mid-term exacerbates these future challenges. The problem is two-fold: UAXS are likely to be used as additional capability for friendly forces, but they are also to be considered as possible threat. The challenges that autonomous systems pose depend not only on their physical vulnerability or the security of the networks and data, but also on the interaction with traditional troops and operating procedures to be followed, their levels of autonomy and Command and Control (C2) chain behind their employment. An extensive experimentation is necessary to address some of these challenges in a Concept Development and Experimentation (CD&E) process aimed to envision solutions. Modelling and Simulation (M&S) tools and architecture are essential to speed up this process in a cost effective way [2, 3]. This paper presents how the development of a particular air operational scenario in an urban environment, according to the Simulation Interoperability Standard Organization (SISO) guide [4] using M&S tools, can support the capability development process for future UAXS as well as their countermeasures, in term of platforms, doctrine and procedures.

So, in the first section the Scenario Development is illustrated, introducing at first the existing preconditions to be considered in order to deal with air operational scenarios in urban environment, such as the challenges to be faced, the definition of the levels of autonomy and interoperability of the UAXS, together with some future capabilities envisioned. Finally, following the Distributed Simulation Engineering and Execution Process (DSEEP) [5] approach followed by SISO Guideline on Scenario Development for Simulation Environments (GSD) [4], the designed operational and conceptual scenarios are described, focusing on the challenges and countermeasures for air domain operations, concerning specifically facing an hostile swarm of Unmanned Autonomous Aerial Vehicles (UAAVs) attacking a critical infrastructure in a highly populated city.

In the second section the designed experimental environment is presented. In particular, the Distributed Simulation Environment under development is based on a federation of simulators interacting with real C2 systems, with a specific focus on the implementation of behavior algorithms for UAXS and message exchange using standards-based C2-Simulation (C2Sim) Interoperability language. Several technologies were exploited such as High Level Architecture (HLA) [6], for simulation Run-Time Infrastructure; Artificial Intelligence (AI) ad-hoc developed modules, for robotic behavior; C2Sim language [7], for interoperation between simulators and C2 systems. For the terrain generation, the part of the mega-city model from the Allied Command for Transformation (ACT) Urbanization Project is being used.

In conclusion, this paper presents the development of an operational scenario of hostile autonomous systems operating in highly built-up and populated environment and representing a threat for a critical infrastructure. Also the Distributed Simulation Environment being developed for executing this scenario is illustrated. All this constitutes an important M&S tool and methodology for concept development activity aimed to

explore the future challenges, which UAxS operations represent in terms of platforms' development and countermeasures to be developed.

2 Scenario Development

2.1 Initial Assumptions

Before starting illustrating the scenario development, some important assumptions needed to be done. Firstly, since this paper focuses on facing aerial autonomous platforms on an urban environment, the main challenges to be faced on such environment in the air domain are to be highlighted. Then the Urbanization Project [8] of the NATO Allied Command for Transformation (ACT) was considered to construct the mega-city environment. Secondly, some peculiarities of the autonomous platforms need to be highlighted, before building the scenario's entities: in particular, the levels of interoperability and of autonomy. For the first, the STANAG 4586 [9] was considered, while, for the last, the documentation from the Counter-UAxS project [10] of the NATO ACT was taken as reference.

2.1.1 City Environment Challenges

NATO doctrine has to ask, how a force can defend itself in a sustainable way without occupying a city for a long period of time and contributing to more problems than it addresses. The ability to deploy a light footprint, disperse and be more agile is fundamental. NATO has to be able to dominate in time and space in all domains, creating temporary corridors and spots: a fluid, undisrupted mobility from the afloat base to the location of effect and back, absorbing damage while dominating a position with all means in all domains. The requirement to ensure adequate and integrated protection throughout the urban and urban littoral battlespace and a single, shared operating picture and concept for all elements of the force is no longer optional, but rather an essential function of these kinds of expeditionary operations.

Force protection will be a major concern in a future urban environment due to the multi-dimensional nature of threats and the ever present risk of escalation. Heavy force protection would have the negative impact of increasing the footprint of the force, decreasing maneuverability, and limiting interaction with the local populace. Ways will have to be developed to balance the need to protect the force while remaining able to accomplish the objectives of the mission. The military should aim to have as small a footprint as possible inside a city. Given the size of most cities, NATO can never expect to be able to build up overwhelming force in terms of mass due to the large number of civilians and other actors. Large cities will quickly 'swallow' and disperse military troops. A key ability for NATO will be to quickly aggregate and disaggregate. Tactics and technology may help to reduce the mission footprint.

The two dominant characteristics affecting aviation urban operations are the existence of manmade construction and the difficulty in distinguishing combatants from non-combatants. These operations may be conducted on or against objectives on a complex urban topology and its adjacent natural terrain. The compressed battlespace in the urban

environment creates unique considerations for planning and conducting aviation operations. These include a comprehensive list of number of considerations that should be taken into account when conducting aviation operations in general, but in case of urban environment, their importance is even greater (e.g. the presence of non-combatants, political considerations that restrict the rules of engagement (ROE), air-to-ground coordination is contributing to the success of urban operations, joint airspace control, future employment of military and commercial UAV will significantly complicate airspace control, centralized integrated air defence systems vs localized threats in the urban environment, ...)

Those, that could be considered as unique for populous areas, are as follows:

- a. Deconfliction in confined airspace - operations in urban canyons, the vertical battlespace, significant risk of collateral damage and friendly fire.
- b. Current NATO Joint Doctrine does not address control of unmanned aerial systems within a high-density airspace control zone.
- c. The three-dimensional ground environment reaches into low-level airspace potentially used by fixed-wing and rotary-wing aircraft and UASs. The location of major airports in or near urban areas often puts the APOD inside the urban airspace. During operations in urban areas, a number of different types of aircraft may share the airspace—tactical aircraft, airlift assets, rotary-wing aircraft, UAS, and civilian aircraft.
- d. Combined arms air-ground maneuver within cities requires operating simultaneously through multiple, interconnected domains and environments. Combined arms air-ground teams require advanced multi-role ground mobility systems with adequate protection and lethality to rapidly transition to offensive operations in severely restricted mobility environments.
- e. The transition from combat operations to a civil authority or local military and security force should be trained for including bridging of the police gap.
- f. The ROE and desire to minimize collateral damage must be balanced against the need to maintain the ability to deliver precision and suppressive fires through a mix of direct and indirect systems. The use of non-lethal technologies for area denial in the urban environment could provide commanders with greater freedom of action while minimizing casualties.
- g. The ability, technically and procedurally, to operate in a GPS denied environment is questionable and should be addressed as part of the force's preparation for operations in the urban environment.

2.1.2 Mega-City “Archaria” Model of ACT Urbanization Project (uP)

In regard of the construction of the environment for the urban scenario, the results of the ACT Urbanization Project (UP) [8] were reused. This UP Project had as goals to deliver a study to the NATO Military Authorities examining potential crises and consequences of urbanization between now and 2035 and their impact on NATO military operations. So, a 2035 metropolitan city 2D/3D model, namely “Archaria”, was created by the Modelling & Simulation Centre of Excellence (M&S CoE) using Geographic Information System (GIS) tools 2D/3D capable (ESRI SW), from which possible

scenarios were extrapolated. The Archaria model took into account that the city should have been littoral and highly populated, with almost 6 million inhabitants, and it was built with layers grouped using the PMESII pattern (Political, Military, Economic, Social, Information and Infrastructure). Some pieces of these layers were reused to build the terrain for the urban scenarios of this paper.

2.1.3 The Level of Interoperability

As defined in STANAG 4586 Edition 3 [9] for NATO UAV there can be identified five levels of interoperability (LoI). The objective of the STANAG 4586 is to specify the interfaces that should be implemented in order to achieve the required LoI between different UAV systems. According to the STANAG the respective operational requirements and concept of operations (CONOPS) defined by NATO countries will determine or drive the required LoI that the specific UAV system will achieve.

The LoI according to the requirements they meet are following:

- Level 1: Indirect receipt and/or transmission of sensor product and associated metadata, for example Key Length Value Metadata Elements from the UAV.
- Level 2: Direct receipt of sensor product data and associated metadata from the UAV. The monitoring is dependent on the type of payload (station) and their number (implemented onboard the UAV) as well as on the type of data format used by the UAV.
- Level 3: Control and monitoring of the UAV payload unless specified as monitor only. The monitor only or control and monitoring is dependent on the type of payload (station), their number and the data format used by the UAV.
- Level 4: Control and monitoring of the UAV, unless specified as monitor only, less launch and recovery. The monitor only or control and monitoring is not affected by the payloads onboard the UAV.
- Level 5: Control and monitoring of UAV launch and recovery unless specified as monitor only.

Unless otherwise stated, LoI 3, 4 & 5 assume control and monitor.

2.1.4 The Level of Autonomy

As stated in the Annex A to CUAXS Disruptive Technology Assessment Game (DTAG) Experiment Results and Conclusions [11, 12] seven levels of autonomy for UAXS can be defined. These levels are numbered from 0 to 6 starting from “human controlled” level to fully autonomous system, based on the degree of human interaction while performing their tasks. But the level of autonomy (LoA) does not depend only on the level of technology and does not represent just the relation between the human operator and autonomous system. It describes also the capabilities of the system to tackle the problems connected with the mission complexity or the difficulty of environment. Based on the knowledge of the level of autonomy specific countermeasures can be performed to destroy the UAAV or at least to prevent him from fulfilling a task.

As explication of the levels of autonomy of UAXS defined during the ACT C-UAXS Project [10] some levels are described in Table 1.

Table 1. The levels of autonomy [10]

Level	Operating functionalities
6	Based on its knowledge of a broader environment, the system can initiate automatically a mission The system gathers, filters, and prioritizes data. The system integrates, interprets data and makes predictions. The system performs final ranking. No information is ever displayed to the human. The system executes automatically and does not allow any human interaction
3	The system is used for a specific mission. The system gathers and displays all the information to the human, but it highlights the non-prioritized, relevant information for the user. The system analyses the information to provide data and makes predictions, though the human is responsible for interpretation of the data. Both the human and the system perform ranking tasks but the results from the system are considered prime. The system allows the human a pre-programmed context-dependent time to veto before execution. The human shadows for contingencies
0	The human only gathers and monitors (defined as filtering, prioritizing and understanding) all data. The human analyses all data, predicts and interprets data. The system does not assist in or perform ranking tasks. The human must do it all. The human alone can execute decision

As can be understood from the Table 1, level 0 refers to UAAV platforms which are remotely piloted and are basically remote sensors. Level 3 is a more properly autonomous system which coordinates with the human for determining its tasks, while level 6 is particular, because it refers to UAxS which are fully autonomous, like level 5 not shown, but which can assign missions by itself depending on external conditions, like a non-traditional unit (i.e., insurgent or terrorist), dormant in the environment, which could activate itself.

2.2 Simulation Objectives, Following the [4] and [5]

The main causes for misusing the autonomous systems are their advantages and capabilities to carry a payload in form of a bomb, ammunition or chemical substances and to launch the destructive load consequently in appropriate distance and altitude to ensure the effective fragments or chemicals dispersal. The objectives of such activities are usually highly populated light targets that don't have higher resistance against possible attack like airports, squares, shopping centers, stations or open sport stadiums. As a result there could be a considerable number of victims and as consequential results it can also paralyze the transport, deter the population, discredit the state bodies and ultimately even disrupt the integrity of the state. The intended aim of the simulation is to find the best method(s) of defense against the swarm of Unmanned Autonomous Aerial Vehicles (UAAVs) inside a mega-city of future.

2.3 Application Space

As illustrated in Fig. 1 and according to the GSD [4] following structure is proposed to closely specify the particular application domain (the instance of application space). In this case, the following application domain can be defined:

1. Simulation Application Mode: Concept Development & Experimentation.
2. Capability: Air defense analysis.
3. Military level: Tactical to Operational (or strategic in special cases).
4. Kind of mission: The defense of an area of interest/critical infrastructure (highly populated area, area of national importance, chemical or power plant, etc.).
5. Staff involved: As the simulation is intended for the experimental and analysis purposes, it is presumed to involve especially the personnel dedicated to the issue (Ground Based Air Defense (GBAD) SMEs, M&SEs).

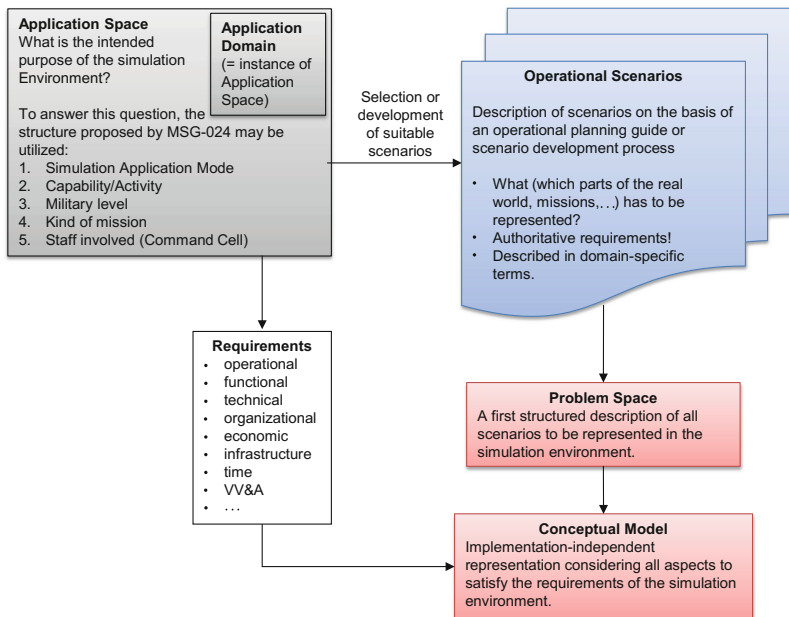


Fig. 1. Role of operational scenario in the early steps of a simulation environment engineering process (SISO GSD PDG, 2016).

2.4 Operational Scenario Development

The simulation will represent the particular scenarios taking place in the urbanized area, the megacity of 2035. On such a place the Critical Energy Infrastructure Protection (CEIP) will be essential and the attention is paid not only to “light targets”. Because according to a continuous development in the area of autonomous systems in the future there could be the possibility of UAAV threat also against more durable critical

infrastructures like power plants, chemical factories or embassy. The amount of possible attacks is considerable and therefore the simulation should be reusable and open to modifications. Moreover, there will be proposed more different scenario variants.

When considering the real world and its impacts, we can assume that the meteorological conditions for the period of an attack will be in all cases ideal. Naturally, the conditions might be unpredictably bad for the unmanned systems to tackle them but we can presume that regarding the future development they will become less or more irrelevant.

According to the political situation we suppose that there is a peacetime or maximally the time of growing crisis. This assumption is connected with the higher security measures in case of war.

Furthermore, we will deal with the possibility of the attack of organized terroristic group instead of an individual isolated attacker. The reason is that even in case of an individual the causalities could be considerable, in case of an organized group the result can be the same or even worse.

When we take into consideration the possible level of technology development of the future city defense as mentioned above in section “Existing Preconditions” it is greatly probable that there will be a highly effective city defense system that will be able to deal with the threats from outside the city (sea or land based). This assumption considers both the detection and subsequent detention or elimination.

According to the Protection Capability Card of the UP [8] the counter UAXS will be capable of detecting, identifying and engaging even small and low altitude UAAV on distances about 1500 m. Furthermore, the Capability Cards will also deal with the questions of operating in GPS denied environment, range of radars netconsidering the obstacles (terrain, buildings), radio visibility and UAS optical sensors around the city.

2.4.1 The Unmanned Autonomous Aerial Vehicles (UAAV)

In order to find the best way of how to protect the specific area of interest against an hostile swarm of UAAVs, first, their functionalities have to be declared along with their possible vulnerabilities. After this classification process the possible weaknesses that were revealed are to be exploited. Obviously, this approach must be preceded by the classification and description of own capabilities.

There exist a number of criteria according to which the Unmanned Autonomous Aerial Systems (UAAS) can be divided. Moreover, there are countless number of systems differing from each other by operating time, maximum speed/altitude/distance, fixed/rotary wing, maneuverability and other capabilities. When it comes to the classification, only one of them as well as a group of parameters can be considered. For the purposes of this work just a several of them, which will be further used in the scenario, considering their application area and size are mentioned.

2.4.2 UAAV Size Category

Regarding the size, the UAAV categories were divided according to the Maximum Gross Takeoff Weight (MGTW) and dimensions as follows:

- Micro UAAV up to 1 kg and up to 10×10 cm.
- Mini UAAV from 1 to 3 kg and up to 30×30 cm.

- Small UAAV from 3 to 10 kg and up to 50×50 cm.
- Medium from 10 to 20 kg and up to 100×100 cm.
- Large more than 20 kg and bigger than one meter in any dimension.

2.4.3 UAAV Application Area

According to the purpose of use and the desired goal of the mission, the UAAV can be divided as follows (considering the particular scenario of this paper):

- Combat UAAV or UACAS (Unmanned Autonomous Combat Aerial Systems) can be characterized by high maneuverability and are able to take part in air or air to ground missions. They usually have shorter operating time but higher travel speed. The demands on these means are usually the communication and cooperation capabilities and usage of weapons (e.g. missiles, high precision bombs).
- Reconnaissance UAAV which contains all the means of Intelligence, Surveillance, Target Acquisition and Reconnaissance (ISTAR) group. They are designed especially for information gathering (terrain/optoelectronic/IR reconnaissance).
- Multipurpose UAASs usually reconnaissance UAAV modified and/or armed. Their primary target is the combat reconnaissance and target acquisition but they can also dispose of missiles.
- There are also UAAV designed for specific purposes like anti-radar UAAV or suicidal systems that represent precision-guided bomb or the cruise missile.

2.4.4 Problem Space

When defining possible scenarios, the probable ways of attack conduction against specific objects are question of their analysis and a complex of all related factors, which is called Intelligence Preparation of Battlefield (IPB).

The work focuses on particular scenario variants concerning UAAV deployment against non-military objects. For the creation of possible terrorist attack scenarios in the megacity of the future different targets are possible:

1. Shopping center – supposed to be a typical appropriate target for an attack due to its huge concentration of people on relatively small area.
2. Critical infrastructure
 - a. Power plant/Chemical plant
 - b. Airport
 - c. Railway (metro) station
 - d. Dam reservoir
3. Embassy.

The following scenarios assume the use of UAAV of sizes from micro to mini. The reason for this condition is that it is easy to buy them and they are easily portable by the means of an individual or by a car. All the UAAV are able to achieve the LoI 3 or higher. According to the kind and phase of the mission the UAAV are able to operate at LoA 1, 3 or 6.

2.5 Conceptual Scenario Development

The process of concept development belongs to the second phase of the DSEEP and the conceptual model embraces the existing UP model and above stated scenarios [13, 14].

2.5.1 Textual Description of the Conceptual Scenario

Example of 1st Scenario Variant

A typical configuration of big shopping centers is a large scale area with three or more floors. There are lots of hypermarkets, small shops, restaurants and other facilities inside the building. There can be even a gas station a few meters from the center. As usual, there is one supply road which cannot be used by ordinary people and just several customer driveways and departure roads.

A hypothetical attack could be led by following minimal number of UAAVs to successfully accomplish their mission:

- I. 1 to 3 UAAVs carrying weapons with fragmentation effect can be guided to the place of a hypermarket through its delicate roof and can be detonated even before falling to ground. As a result there could be primarily great number of victims which can lead to a panic, attempts to escape the place and then there could appear other victims as secondary effect.
- II. 1 to 2 multipurpose UAAVs with poisonous substances (or just any aerosol to evoke a panic and chaos) can either spread it round the park place or inside the building. When not considering the direct victims other negative unpredictable actions of individuals cannot be excluded causing secondarily other victims.
- III. 1 multipurpose UAAV with precision guidance and cumulatively/inflammable weapon can be guided to a gas station.

A roadblock on emergency escape road caused by a fire or its destruction will lead to further dominoes effect in panic and number of victims.

Example of 2nd Scenario Variant

Three Swarms of enemy/terrorist UAAVs each consisting of 5 vehicles is intended to attack the chemical plant. The three UAAVs from each swarm are special suicide UAAVs representing precision guided bombs, the other two vehicles are fully autonomous UACASs designed to cooperate and protect the suicides to allow them fulfilling the task. One vehicle is the reconnaissance ISTAR UAAVs for target acquisition and information gathering. Then there are two multipurpose UAAVs equipped with missiles. Their goal is to block the road, prevent the people from escaping and consequently to cause other domino effect increasing the panic and loss of life.

The location of individual swarms of UAAVs and their primary preprogrammed routes are illustrated in Fig. 2. Moreover, for the escape from the launch site the aggressors have several minutes, which is enough for merging with the rest of the population. The time and population can be calculated by the UP project application. The population up to the distance of 1.5 km from the airport counts more than 100 000 people.

The static autonomous defense system with visual and electromagnetic detection is placed according to the Fig. 2. The radar is in surveillance mode. When the UAASs are detected (by the visual or electromagnetic means) the weapon system is activated and

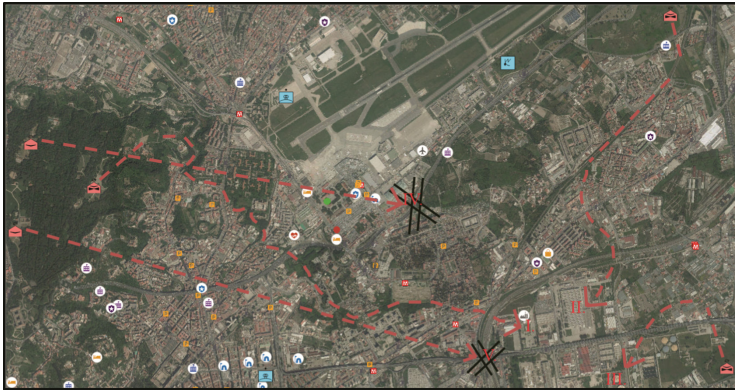


Fig. 2. The starting positions and routes of 2nd scenario variant (created with Urbanization Project Archaria model Decision Support tool).

the process of further detection follows. When the target is identified as hostile an appropriate counter action kinetic or non-kinetic is proposed taking into consideration the possible collateral damages and loss of life. When looking for the best counter-measure, the detection of level of autonomy is also performed.

The final phase of the target elimination by using the weapons is not autonomous and is up to the responsible person. When the person orders a fire command the weapon system launches the direct energy against the enemy. The defense system then evaluates and reports the results to the responsible person for consecutive action.

Termination Conditions

Termination condition: The fire was successful and hits all the targets or the enemy swarm fulfilled the task. Specified time elapsed

3 Distributed Simulation Environment (DSE)

The air operational scenario illustrated in this paper is intended to be executed in a Distributed Simulation Environment (DSE) under development at the Modelling and Simulation Centre of Excellence (M&S CoE).

The architecture of the DSE is designed to be a standards-based federation of simulators interoperable with real C2 systems, scalable and continuously integrable with other elements in following evolutions. The importance of an hybrid experimental framework as a key instrument to operationalize autonomous systems' implementation in the current and future missions is well discussed in [15]. The simulators' federation is based on an High Level Architecture (HLA) Run-Time Infrastructure (RTI) [6], since this standard is well recognized for its maturity and suitable as experimental backbone for autonomous systems' simulation [16]. The HLA federates of the DSE are constructive and virtual simulators, responsible for the scenario management, including the terrain generation, entity, event interaction and robot simulations in the scenario, and a

Communications and Networks simulator, for the communications among platforms in the scenario with the added capability to be stimulated by pretending cyber-attacks. The kinetic and cooperative behavior of the UAAVs during the flight is implemented with add-on modules developed in Simulink®, with the possibility to compile the code as plug-in for the simulator. Another federate of the HLA RTI is the LVC Gateway, which acts as an interface between simulators and real C2 systems, performing filtering and message translations between HLA or DIS [17] to appropriate formats for C2 systems, such as MIP [18] and NFFI [19], in order to display the overall Comprehensive Operational Picture (COP) or part of it (like displaying only land/air/naval units).

The aim is to realize mission control and feedback during a simulation involving UAxS, using standard-based languages and data model which can be used in the military and civilian worlds and that are as close as possible to the human natural language in which standard military orders are issued. For this reason, this distributed simulation environment is designed to be used as test bed for C2Sim language (NATO STO NMSG-145, 2017) experimentation and extension of its Logical Data Model (LDM) to Autonomous Systems. By the moment being, only some elements of the architecture support the use of C-BML [20] and MSDL [21] languages, which are the basis of the C2Sim, in order to exchange messages (orders and reports) during the scenario execution.

In particular, this architecture, depicted in Fig. 3, has the following components:

- NATO and National C2C4ISR systems;
- LVC Gateway;
- MASA SWORD Simulator with AI add-on modules and its BML Connector;
- VT MÄK VR-FORCES Simulator;
- Communication, Network and Cyber Simulator;
- Artificial Intelligence (AI) Kinetic behavior models (built in Simulink®) for VT MÄK VR-FORCES Simulator;
- HLA Run-Time Infrastructure (RTI);
- Experimental C2Sim interface for VT MÄK VR-FORCES Simulator;
- C2Sim Server.

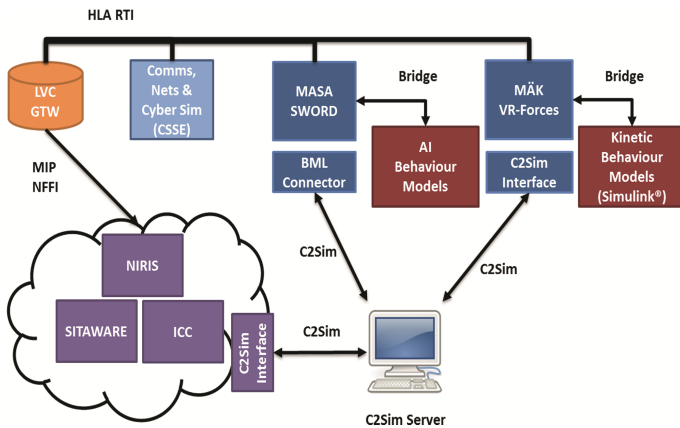


Fig. 3. Logical architecture of the distributed simulation environment.

3.1 Urban Terrain Generation

Concerning the terrain generation, as introduced in the first section, some pieces of layers of the Archaria mega-city model were imported in the form of Shape (.shp) files in the simulators of the DSE. The lessons learned during the building of the model of the future megacity were taken into consideration [22]. An example of the layers forming the urban model is shown in Fig. 4.

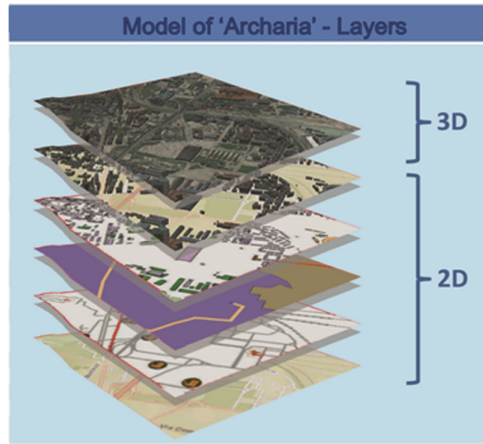


Fig. 4. Layers of the Archaria Model – ACT Urbanization Project.

4 Way Ahead

The overall idea for the follow-on activity regards both the scenario development and the DSE evolution. Concerning the scenario, it is intended to create a tool for the multi-domain defense concept development by connecting (extending) this scenario with other domains. The future possible scenarios can also consider an enemy attack of bigger military UACASs from higher altitudes above the maximum range of city self-defense (possible “Scenario variant” depending on the city defense technologies) or we can limit the level of autonomy.

Regarding the evolution of the DSE, the idea is to include in the future virtual UAVs and a real UAV platforms in the DSE for control and situation awareness, both interacting with real C2 systems. The Robotic Operating System (ROS) [23] is a good candidate for the middleware which can allow messages exchange among virtual robots and between virtual and real platforms. The real platforms domain of the architecture should be composed by prototypes or on-the-market UAV systems and their controlling subsystems, mainly the control station supporting STANAG 4586 and a Data Distribution Service (DDS) [24] Bus for all communications. Moreover, further work is needed to extend the functionalities of the C2Sim interoperability languages, especially regarding the evolution of C2Sim interface for simulators and C2 systems to support message exchange involving autonomous systems.

5 Conclusions

In conclusion, this paper presents the development of an air operational scenario of hostile autonomous systems operating in highly built-up and populated environment and representing a threat for a critical infrastructure. The preconditioning assumptions regarding the environment and the autonomous systems of interest were illustrated: the peculiarities of a urban environment for air domain operations and those regarding unmanned autonomous platforms were highlighted. These are fundamental to build operational scenarios able to stimulate the concept development activity in the right direction. Therefore, the steps necessary to develop such scenario according to the SISO GSD [4] are described up to the textual description of the conceptual scenario. Finally the real experimentation environment being developed for executing this scenario is illustrated. State-of-the art and under-development standards are the focus of the architecture of this DSE. All this constitutes an important M&S tool and methodology for concept development activity aimed to explore the future challenges, which UAxS operations represent in terms of platforms' development and countermeasures to be developed.

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